Retail Order API

1.0.0 OAS 3.0

/api/schema/

Detailed API for managing orders, items, and sales analytics.

Authorize

api

GET /api/customers/{customer_id}/orders/ Orders by Customer

Retrieve a list of all orders placed by a specific Customer ID.

Parameters

Try it out

Name	Description
customer_id * required string (path)	customer_id

Responses

Code	Description	Links
200		*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
[
  {
    "invoice_no": "string",
    "customer_id": "string",
    "invoice_date": "2026-03-13T20:46:16.904Z",
    "country": "string",
    "order_status": "string",
    "total_value": "string",
    "total_items_count": "string",
    "items": [
      {
        "id": 0,
        "stock_code": "string",
        "description": "string",
        "quantity": 9223372036854776000,
        "price": "-2487.4",
        "line_total": "string"
      }
    ]
  }
]
```


404 Customer not found or has no orders. *No links*

Media type

application/json

Example Value Schema


```
{
  "message": "No orders found for this customer."
}
```


GET **/api/orders/** List all Orders

Retrieve a list of all orders in the system.

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Media type application/json Controls Accept header. **Example Value** Schema	*No links*


```
[
  {
    "invoice_no": "string",
    "customer_id": "string",
    "invoice_date": "2026-03-13T20:46:16.905Z",
    "country": "string",
    "order_status": "string",
    "total_value": "string",
    "total_items_count": "string",
    "items": [
      {
        "id": 0,
        "stock_code": "string",
        "description": "string",
        "quantity": 9223372036854776000,
        "price": "",
        "line_total": "string"
      }
    ]
  }
]
```


POST /api/orders/ Create a New Order

Create a new order along with its nested line items.

Parameters

Try it out

No parameters

Responses

Code	Description	Links
201	Order successfully created.	*No links*

Media type

application/json

Controls `Accept` header.

Examples

Created Order Response

Example Value Schema


```
{
  "invoice_no": "536367",
  "customer_id": "12345",
  "invoice_date": "2026-03-13T14:30:00Z",
  "country": "United Kingdom",
  "order_status": "Pending",
  "total_value": 13.52,
  "total_items_count": 8,
  "items": [
    {
      "stock_code": "84879",
      "description": "ASSORTED COLOUR BIRD ORNAMENT",
      "quantity": 8,
      "price": "1.69",
      "line_total": 13.52
    }
  ]
}
```


400	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "invoice_no": [
    "Order with this invoice no already exists."
  ]
}
```

GET /api/orders/{id}/ Get Order Details

Fetch a specific order by its unique invoice number, including all nested line items.

Parameters

Try it out

Name	Description
id * required string (path)	

Responses

Code	Description	Links
200	Successful retrieval of the order. Media type: application/json Controls Accept header. Examples: Example Order Response Example Value Schema	No links


```
{
  "invoice_no": "536365",
  "customer_id": "17850",
  "invoice_date": "2026-03-13T12:00:00Z",
  "country": "United Kingdom",
  "order_status": "Pending",
  "total_value": 150.25,
  "total_items_count": 5,
  "items": [
    {
      "stock_code": "85123A",
      "description": "WHITE HANGING HEART",
      "quantity": 5,
      "price": "30.05",
      "line_total": 150.25
    }
  ]
```

```
}
```

404	Order not found.	*No links*

Media type

application/json

Example Value Schema

```
{
  "error": "Order not found"
}
```

PATCH /api/orders/{id}/ Update Order

Partially update an order. You only need to send the fields you want to change (e.g., updating the `order_status` to 'Shipped').

Parameters

Try it out

Name	Description
id * required `string` *(path)*	id

Responses

Code	Description	Links
200	The order was successfully updated.	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema

```
{
  "invoice_no": "string",
  "customer_id": "string",
  "invoice_date": "2026-03-13T20:46:16.906Z",
  "country": "string",
  "order_status": "string",
  "total_value": "string",
  "total_items_count": "string",
  "items": [
    {
      "id": 0,
      "stock_code": "string",
      "description": "string",
      "quantity": 9223372036854776000,
      "price": "0043.6",
      "line_total": "string"
    }
  ]
}
```


400 Bad Request (e.g., invalid data types). *No links*

Media type

application/json

Example Value Schema


```
{
  "order_status": [
    "Invalid status."
  ]
}
```


404 Order not found. *No links*

Media type

application/json

Example Value Schema


```
{
  "error": "Order not found"
}
```


DELETE **/api/orders/{id}/** Delete Order

Permanently delete an order. This will also delete all associated items from the database.

Parameters

Try it out

Name	Description
id * required string (path)	id

Responses

Code	Description	Links
204	Successfully deleted (No content returned).	*No links*

404	Order not found.
	Media type
	application/json
	Example Value Schema
	{
	"error": "Order not found"
	}
		No links

GET /api/orders/analytics/advanced/ Advanced Sales Analytics

Calculates total revenue, identifies top 5 selling products, and provides country-wise distribution.

Parameters

Try it out

No parameters

Responses

Code	Description	Links
200	Analytics Success	*No links*

Media type

application/json

Controls `Accept` header.

Examples

Analytics Example

Example Value Schema


```
{
  "total_revenue": 15420.5,
  "top_5_products": [
    {
      "stock_code": "85123A",
      "description": "WHITE HANGING HEART",
      "total_sold": 500
    }
  ],
  "country_distribution": [
    {
      "country": "United Kingdom",
      "order_count": 150
    }
  ]
}
```


GET /api/orders/analytics/summary/ Basic Order Analytics

Returns total tracked orders, items in the stockroom, and a breakdown of orders by status.

Parameters

Try it out

No parameters

Responses

Code	Description	Links
200	Media type: application/json Controls Accept header. Examples: Basic Metrics Example Value Schema	No links

```
{
  "metrics": {
    "total_orders_tracked": 120,
    "physical_items_in_stockroom": 450
  },
  "status_breakdown": {
    "Pending": 15,
    "Collected": 105
  }
}
```

GET /api/orders/date/{date_str}/ Filter Orders by Date

Retrieve all orders placed on a specific date.

Parameters

Try it out

Name	Description
date_str * required string (path)	Format: YYYY-MM-DD date_str

Responses

Code	Description	Links
200		No links

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
[
  {
    "invoice_no": "string",
    "customer_id": "string",
    "invoice_date": "2026-03-13T20:46:16.907Z",
    "country": "string",
    "order_status": "string",
    "total_value": "string",
    "total_items_count": "string",
    "items": [
      {
        "id": 0,
        "stock_code": "string",
        "description": "string",
        "quantity": 9223372036854776000,
        "price": "-",
        "line_total": "string"
      }
    ]
  }
]
```


400 Invalid date format. *No links*

Media type

application/json

Example Value Schema


```
{
  "error": "Invalid date format. Please use YYYY-MM-DD."
}
```


GET **/api/orders/large/** Filter Large Orders

Returns orders where the sum of all item quantities meets or exceeds the threshold.

Parameters Try it out

Name	Description
threshold **integer** *(query)*	Minimum quantity of items to filter by threshold

Responses

Code	Description	Links
200	Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*


```
[
  {
    "invoice_no": "string",
    "customer_id": "string",
    "invoice_date": "2026-03-13T20:46:16.908Z",
    "country": "string",
    "order_status": "string",
    "total_value": "string",
    "total_items_count": "string",
    "items": [
      {
        "id": 0,
        "stock_code": "string",
        "description": "string",
        "quantity": 9223372036854776000,
        "price": "-2343",
        "line_total": "string"
      }
    ]
  }
]
```


Schemas

Order {

invoice_no*
string
maxLength: 50
The unique invoice number identifying this specific order.

customer_id
string
nullable: true
maxLength: 50
The unique identifier for the customer who placed the order.

invoice_date*
string($date-time)
The exact date and time the order was placed.

country
string
maxLength: 100
The country where the order was placed or shipped.

order_status
string
maxLength: 50
The current processing status of the order (e.g., Pending, Shipped).

total_value*
string
readOnly: true

total_items_count*
string
readOnly: true

items*
[OrderItem {

id*
integer
readOnly: true

stock_code*
string
maxLength: 50
The unique stock keeping unit (SKU) or product code.

description
string
nullable: true
maxLength: 255
A brief text description of the product.

quantity*
integer($int64)
maximum: 9223372036854776000
minimum: -9223372036854776000
The number of units purchased for this specific item.

price*
string($decimal)
pattern: ^-?\d{0,8}(?:\.\d{0,2})?$
The price per single unit of this item.

line_total*
string

```
                                        readOnly: true
                        }]
}
```

OrderItem

```
{
    id*
                        integer
                    readOnly: true
    stock_code*
                        string
                    maxLength: 50
                    The unique stock keeping unit (SKU) or product code.

    description
                        string
                    nullable: true
                    maxLength: 255
                    A brief text description of the product.

    quantity*
                        integer($int64)
                    maximum: 9223372036854776000
                    minimum: -9223372036854776000
                    The number of units purchased for this specific item.

    price*
                        string($decimal)
                    pattern: ^-?\d{0,8}(?:\.\d{0,2})?$
                    The price per single unit of this item.

    line_total*
                        string
                    readOnly: true
}
```